

## FAREWELL SPIT, New Zealand

WMO: 935270

Lat: 40.545S

Lon: 173.001E

Elev: 15

StdP: 14.69

Time Zone: 12.00 (NZL)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 7             | 39.7       | 41.7 | 30.5                          | 24.7 | 47.1 | 33.6 | 28.2 | 45.7 | 36.9                  | 52.5 | 33.3 | 52.5 | 6.2                   | 260  | 0.632 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 2             | 11.2                   | 74.0            | 65.5 | 72.2 | 64.4 | 70.7 | 63.5 | 67.5                | 71.4 | 66.3 | 69.9 | 65.2 | 68.9 | 14.6                 | 270  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 66.0                            | 96.1 | 69.1 | 64.9 | 92.2 | 68.3 | 63.7 | 88.3 | 67.4 | 32.0          | 71.6 | 31.0 | 70.1 | 30.2 | 68.9 | 72.7              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |      |                    | n-Year Return Period Values of Extreme Temperature |           |      |            |      |            |      |            |      |
|-------------------|------|------|----------------------------|------|------|--------------------|----------------------------------------------------|-----------|------|------------|------|------------|------|------------|------|
| 1%                |      | 2.5% | 5%                         | Mean |      | Standard Deviation |                                                    | n=5 years |      | n=10 years |      | n=20 years |      | n=50 years |      |
| Min               |      | Max  |                            | Min  | Max  |                    | Min                                                | Max       | Min  | Max        | Min  | Max        | Min  | Max        |      |
| (a)               | (b)  | (c)  |                            | (e)  | (f)  | (g)                | (h)                                                | (i)       | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 32.8              | 29.2 | 26.6 | DB                         | 37.4 | 77.8 | 2.8                | 2.1                                                | 35.3      | 79.3 | 33.7       | 80.5 | 32.1       | 81.6 | 30.0       | 83.1 |
|                   |      |      | WB                         | 34.8 | 69.3 | 1.9                | 1.5                                                | 33.4      | 70.4 | 32.3       | 71.2 | 31.2       | 72.0 | 29.9       | 73.1 |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 57.3   | 64.0  | 64.8  | 62.8  | 59.0  | 55.6  | 51.7  | 50.2  | 51.1  | 53.4  | 55.6  | 58.2  | 61.5 |
|                                                                                  | DBStd       | 5.80   | 3.19  | 2.85  | 3.01  | 2.98  | 3.37  | 3.50  | 3.06  | 2.88  | 2.70  | 2.74  | 2.97  | 3.09 |
|                                                                                  | HDD50       | 89     | 0     | 0     | 0     | 0     | 3     | 23    | 35    | 22    | 5     | 1     | 0     | 0    |
|                                                                                  | HDD65       | 2890   | 56    | 35    | 81    | 179   | 290   | 400   | 459   | 430   | 348   | 292   | 206   | 114  |
|                                                                                  | CDD50       | 2746   | 435   | 414   | 396   | 271   | 177   | 72    | 41    | 57    | 107   | 174   | 245   | 357  |
|                                                                                  | CDD65       | 74     | 26    | 29    | 12    | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 1     | 6    |
|                                                                                  | CDH74       | 55     | 28    | 18    | 3     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 6    |
|                                                                                  | CDH80       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
| Wind                                                                             | WSAvg       | 13.0   | 13.5  | 12.2  | 12.1  | 11.5  | 12.3  | 12.2  | 12.3  | 12.1  | 13.9  | 15.0  | 15.4  | 13.8 |
| Precipitation                                                                    | PrecAvg     | 60.2   | 4.3   | 4.9   | 4.1   | 5.4   | 4.9   | 5.6   | 5.7   | 6.3   | 5.1   | 5.5   | 4.7   | 4.3  |
|                                                                                  | PrecMax     | 90.6   | 7.9   | 14.5  | 6.7   | 10.9  | 10.4  | 9.6   | 11.8  | 14.1  | 10.7  | 11.4  | 9.8   | 11.4 |
|                                                                                  | PrecMin     | 42.0   | 1.3   | 1.7   | 1.7   | 0.8   | 0.3   | 2.0   | 1.7   | 2.4   | 1.5   | 2.0   | 1.2   | 1.6  |
|                                                                                  | PrecStd     | 11.6   | 1.9   | 3.2   | 1.5   | 3.0   | 2.3   | 2.1   | 2.5   | 2.3   | 2.2   | 2.4   | 2.6   | 2.4  |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 77.1  | 76.3  | 73.8  | 68.9  | 65.4  | 61.2  | 59.2  | 60.2  | 63.1  | 66.5  | 71.1  | 74.3 |
|                                                                                  |             | MCWB   | 67.2  | 66.9  | 65.4  | 62.2  | 60.4  | 57.6  | 55.2  | 54.9  | 56.1  | 58.4  | 61.3  | 64.6 |
|                                                                                  | 2%          | DB     | 74.0  | 73.9  | 71.5  | 66.9  | 63.6  | 59.4  | 57.9  | 58.6  | 61.0  | 64.1  | 67.7  | 71.0 |
|                                                                                  |             | MCWB   | 65.6  | 65.7  | 64.1  | 60.9  | 59.4  | 56.3  | 54.3  | 53.9  | 55.4  | 57.3  | 59.6  | 63.0 |
|                                                                                  | 5%          | DB     | 72.0  | 72.3  | 70.0  | 65.6  | 62.2  | 58.4  | 56.7  | 57.4  | 59.7  | 62.5  | 65.7  | 69.0 |
|                                                                                  |             | MCWB   | 64.3  | 64.7  | 63.1  | 60.0  | 58.4  | 55.4  | 53.3  | 53.2  | 54.7  | 56.3  | 58.5  | 61.9 |
|                                                                                  | 10%         | DB     | 70.2  | 70.8  | 68.5  | 64.3  | 61.1  | 57.4  | 55.5  | 56.2  | 58.4  | 61.0  | 64.1  | 67.4 |
|                                                                                  |             | MCWB   | 63.4  | 63.7  | 62.1  | 59.4  | 57.4  | 54.4  | 52.1  | 52.5  | 54.0  | 55.4  | 57.5  | 60.9 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 69.3  | 69.3  | 67.7  | 64.9  | 62.9  | 59.2  | 57.3  | 57.1  | 58.4  | 60.7  | 63.3  | 66.6 |
|                                                                                  |             | MCDB   | 74.9  | 73.5  | 71.4  | 66.6  | 63.8  | 59.9  | 57.9  | 58.4  | 60.3  | 63.3  | 67.9  | 71.4 |
|                                                                                  | 2%          | WB     | 67.4  | 67.6  | 66.1  | 63.2  | 61.1  | 57.6  | 55.5  | 55.6  | 57.3  | 59.1  | 61.5  | 64.8 |
|                                                                                  |             | MCDB   | 71.7  | 71.2  | 69.3  | 65.1  | 62.5  | 58.8  | 56.8  | 57.3  | 59.1  | 61.7  | 65.4  | 68.5 |
|                                                                                  | 5%          | WB     | 66.1  | 66.5  | 64.8  | 61.9  | 59.8  | 56.4  | 54.0  | 54.4  | 56.4  | 58.0  | 60.3  | 63.7 |
|                                                                                  |             | MCDB   | 69.7  | 70.2  | 68.0  | 63.9  | 61.5  | 58.0  | 55.8  | 56.4  | 58.4  | 60.5  | 63.7  | 67.0 |
|                                                                                  | 10%         | WB     | 64.9  | 65.3  | 63.5  | 60.8  | 58.3  | 55.0  | 52.5  | 53.2  | 55.4  | 56.9  | 59.1  | 62.5 |
|                                                                                  |             | MCDB   | 68.7  | 69.2  | 66.9  | 63.0  | 60.4  | 56.9  | 54.7  | 55.5  | 57.6  | 59.5  | 62.1  | 65.6 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 11.2  | 11.2  | 11.0  | 10.2  | 9.4   | 9.2   | 9.4   | 9.9   | 9.9   | 10.5  | 10.9  | 10.6 |
|                                                                                  |             | MCDBR  | 13.2  | 12.8  | 12.2  | 11.2  | 10.1  | 9.0   | 9.7   | 10.4  | 10.9  | 12.1  | 12.6  | 12.4 |
|                                                                                  |             | MCWBR  | 7.3   | 7.3   | 7.0   | 7.3   | 7.2   | 7.1   | 7.4   | 7.4   | 7.4   | 7.9   | 7.3   | 7.4  |
|                                                                                  | 5% WB       | MCDBR  | 10.8  | 10.5  | 10.1  | 8.4   | 7.8   | 7.0   | 8.3   | 8.7   | 9.1   | 9.6   | 10.0  | 10.0 |
|                                                                                  |             | MCWBR  | 6.9   | 7.0   | 6.9   | 6.7   | 6.8   | 6.5   | 7.3   | 7.2   | 7.3   | 7.5   | 6.9   | 6.9  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.345  | 0.336 | 0.322 | 0.314 | 0.302 | 0.301 | 0.294 | 0.304 | 0.333 | 0.338 | 0.332 | 0.341 |      |
|                                                                                  | taud        | 2.576  | 2.621 | 2.661 | 2.666 | 2.660 | 2.634 | 2.656 | 2.643 | 2.539 | 2.545 | 2.586 | 2.595 |      |
|                                                                                  | Ebn at Noon | 312    | 307   | 298   | 280   | 263   | 252   | 264   | 279   | 289   | 303   | 314   | 315   |      |
|                                                                                  | Edh at Noon | 33     | 30    | 26    | 23    | 20    | 18    | 19    | 23    | 29    | 32    | 32    | 33    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 2187   | 1922  | 1502  | 1001  | 684   | 520   | 598   | 855   | 1230  | 1643  | 2037  | 2120  |      |
|                                                                                  | RadStd      | 145    | 154   | 70    | 88    | 30    | 33    | 39    | 47    | 78    | 129   | 158   | 156   |      |

## Historical Trends

|      |                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------|------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|      |                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
| (46) | Station Only           | +0.81 | +2.38   | N/A    | +0.65   | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) | Regional (6 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page